be some that interact with the rest of the universe *precisely* with those variables that found themselves having a particular value in the past.

It is hardly surprising that there are "special" subsets in a universe as vast as ours. It is not surprising that *someone* wins the lottery: someone wins it every week. It is unnatural to assume that the entire universe has been in an incredibly "special" configuration in the past, but there is nothing unnatural in imagining that the universe has parts that are "special."

If a subset of the universe is special in this sense, then for this subset the entropy of the universe is low in the past, the second law of thermodynamics obtains; memories exist, traces are left—and there can be evolution, life, and thought.

In other words, if in the universe there is something like this—and it seems natural to me that there could be—then we belong to that something. Here, "we" refers to that collection of physical variables to which we commonly have access and by means of which we describe the universe. Perhaps, therefore, the flow of time is not a characteristic of the universe: like the rotation of the heavens, it is due to the particular perspective that we have from our corner of it.

But why should *we* belong to one of *these* special systems? For the same reason that apples grow in northern Europe, where people drink cider, and grapes grow in the south, where people drink wine; or that I was born where people happen to speak my native language; or that the sun which warms us is at the right distance from us—not too close and not too far away. In all these cases, the "strange" coincidence arises from confusing the causal relations: it isn't that apples grow where people drink cider, it is that people drink cider where apples grow. Put this way, there is no longer anything strange about it.

Similarly, in the boundless variety of the universe, it may happen that there are physical systems that interact with the rest of the world through those particular variables that define an initial low entropy. With regard to *these* systems, entropy is constantly increasing. There, and not elsewhere, there are the typical phenomena associated with the flowing of time: life is possible, together with evolution, thought, and our awareness of time passing. There, the apples grow that produce our cider: time. That sweet juice that contains all the ambrosia and all the gall of life.

INDEXICALITY

When we do science, we want to describe the world in the most objective way possible. We try to eliminate distortions and optical illusions deriving from